

Maryam Amjad

AI/ML Engineer | LLM & RAG Systems | Deep Learning | Computer Vision | NLP
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PROFESSIONAL SUMMARY

AI/ML Engineer specializing in production-grade, LLM-powered systems – including real-time prediction pipelines, Retrieval-Augmented Generation (RAG) architectures, knowledge-graph-based query engines, and multi-modal healthcare AI. Strong foundation in Python, Deep Learning, Computer Vision, and NLP, with hands-on experience across the full machine learning lifecycle: data pipeline engineering, model integration, evaluation, and cloud deployment on AWS.

PROFESSIONAL EXPERIENCE

Junior Machine Learning Engineer

Tensor Labs

Sep 2025 – Present

- Real-time ML prediction pipelines:** Designed and deployed real-time machine learning pipelines to detect breakout patterns across multiple cryptocurrency markets (Ethereum, Dogecoin, Solana), processing streaming market data at 5-minute intervals.
- RAG-based query engine (DiscoverIQ):** Engineered a Retrieval-Augmented Generation (RAG) pipeline that translates natural language into SQL, integrating a Neo4j knowledge graph, Qdrant vector search, and LLM-based query generation over PostgreSQL.
- Multi-modal clinical AI assistant (MedGemma):** Built core modules for a multi-modal clinical assistant supporting medical image diagnostics, prescription OCR, and drug-interaction checks using vision-language and LLM models.
- Data pipeline reliability:** Diagnosed and resolved data pipeline issues – timestamp/interval synchronization and signal threshold tuning – to improve prediction reliability and system uptime.
- Technical workshops:** Delivered internal workshops on Mem0 (LLM memory systems) and Harness Engineering for AI agent development; authored an article on Harness Engineering and Open Knowledge Format (Medium, forthcoming).

Artificial Intelligence Intern

AIM Lab, Islamabad

Jun 2023 – Aug 2023

- AI-powered web application:** Built a tool that auto-generates presentation slide decks from natural language prompts using NLP and computer vision techniques.
- Transfer learning:** Applied pre-trained vision models for content understanding and image matching tasks.

Lab Demonstrator – Machine Learning

FAST NUCES Islamabad

Sep 2024 – Dec 2024

- ML instruction:** Led weekly lab sessions on feature engineering, hyperparameter tuning, and applied machine learning demonstrations for undergraduate students.

KEY PROJECTS

Eth Breakout – Real-Time Crypto Prediction System:

Real-time breakout prediction system for Ethereum, Dogecoin, and Solana using rule-based and machine learning signal generation across 5-minute market candles; includes a liquidation/liquidity heatmap module visualizing leveraged position risk from open interest and funding-rate data. *Technologies:* Python, SQLAlchemy, Pandas, Scikit-learn, PostgreSQL, WebSockets, Streamlit

DiscoverIQ – RAG-Based Text-to-SQL & Knowledge Graph Query Engine:

End-to-end Retrieval-Augmented Generation (RAG) system that converts natural language into executable SQL queries, combining Neo4j knowledge-graph retrieval, Qdrant vector search, and LLM-based query generation for schema-aware SQL execution.

Technologies: Python, Neo4j, Qdrant, PostgreSQL, LangChain, Groq LLM

MedGemma – Multi-Modal Clinical AI Assistant:

Full-stack, multi-modal clinical AI assistant integrating vision-language models for medical image diagnostics, OCR-based prescription reading, and LLM-driven drug-interaction checking, alongside personal health tracking. *Technologies:* MedGemma-4b-it,

Whisper-large-v3, TxGemma-9b-chat, FDA API, Python, Flask

LLM Evaluation Toolkit:

Open-source evaluation toolkit covering 12 LLM evaluation categories (text generation, summarization, RAG, text-to-SQL, safety & robustness, long-context reasoning, multi-modal, and more), with runnable notebooks and an interactive Streamlit dashboard; used as the basis for an internal engineering workshop. *Technologies:* Python, Groq, RAGAS, DeepEval, HuggingFace Evaluate, PromptBench, Streamlit

EDUCATION

B.S. in Artificial Intelligence

FAST NUCES, Islamabad

Relevant Coursework: Deep Learning, Generative AI, Computer Vision, NLP

Jul 2021 – Jun 2025

CORE SKILLS

Programming: Python, C++, C, SQL

ML/DL Frameworks: TensorFlow, PyTorch, Keras, Scikit-learn

LLM & GenAI: LangChain, Groq, Hugging Face, Transformers, RAG (Retrieval-Augmented Generation), Vector Databases, Prompt Engineering, RAGAS, DeepEval, Mem0, Whisper

Data/NLP/CV: Pandas, NumPy, Matplotlib, OpenCV, NLTK, SpaCy, Tesseract

Databases: PostgreSQL, MySQL, Neo4j, Qdrant, SQLAlchemy

MLOps/DevOps: Docker, Kubernetes, Jenkins, Airflow, MLFlow, CI/CD, DVC, Git

Cloud/Other: AWS, Flask, Streamlit, Selenium, Jupyter Notebook

CERTIFICATIONS & AWARDS

AWS Certified AI Practitioner (AWS, Passed March 2026) • **Generative AI with LLMs** (DeepLearning.AI, 2025) • **AI Agents with RAG and LangChain** (IBM, 2025) • **Convolutional Neural Networks** (DeepLearning.AI, 2024) • **AWS Cloud Essentials** (AWS, 2025)